Assignment 1 – Summary

GitHub repository

GitHub username	ishan-monie
Repository name	advanced-ml-hw1
Link to the code	https://github.com/ishan-monie/advanced-ml-hw1

All of the code used for this assignment is in this repository.

How to run the code

Step 1. Download the repository (green *Code* button, then *Download ZIP*) or clone it, and install the Python packages:

```
pip install numpy matplotlib
```

Step 2. Shuttle problem. Open a terminal in the `q1_shuttle` folder and run:

```
python run_q1.py     # solves the problem 3 ways and saves the 3 plots in figures/
python test_q1.py    # runs the 23 edge-case and correctness checks (PASS/FAIL for each)
```

The README in the repository repeats these instructions and explains what each file does. Instructions for the policy gradient questions will be added to the same README.

Question 2: Value and Policy Iteration (the shuttle problem)

Each period we choose whether to dispatch a shuttle (cost 100, takes up to 15 people) or wait (every person left waiting costs 2). Between 1 and 5 new people arrive each period. We solve this three ways. A lower cost is better in every graph.

Assumptions

The problem leaves some details open, so these are the choices we made:

- **What we track.** The only thing that matters when deciding is how many people are waiting at the start of a period (0 to 200). We do not track the time of day or what happened before.
- **Costs, not rewards.** Everything is a cost, so we look for the *lowest* total cost. The “value” in the graphs is the expected total cost, so lower is better.
- **Order of events in a period.** We first decide (wait or dispatch), then pay, then the new people arrive.
- **Dispatching.** The shuttle is always available. It takes up to 15 people, or everybody if fewer than 15 are waiting. We allow dispatching with an empty station, even though it is wasteful.
- **The waiting cost.** The cost of 2 per person is charged on the people *left* after the shuttle has gone (everyone, if we wait).
- **Full station.** The station holds at most 200 people. If more would arrive, the extra people are turned away (they are lost, they do not wait). We checked that this choice does not change the answer.
- **Arrivals.** Each period 1, 2, 3, 4 or 5 people arrive, each with chance 1/5, independent of the past.
- **Discounting.** A cost one period later counts 0.95 as much. The cost in the first period is not discounted.
- **The end of the 500 periods (enumeration).** After the last period nothing more is paid, so people still waiting at that point cost nothing extra.
- **The plan.** The best action depends only on the number of people waiting, and it is the same in every period (this is what value iteration and policy iteration find).
- **Stopping and ties.** Value iteration stops when no value changes by more than 1e-10. Policy iteration starts from “never dispatch” and keeps its current action when two actions cost exactly the same.

(a) Enumeration, 500 periods

We start at the end of the 500 periods, where nothing is left to pay, and work backwards, keeping the cheaper of the two actions at every step. The graph shows the best expected total cost at time 0 for each number of people waiting.

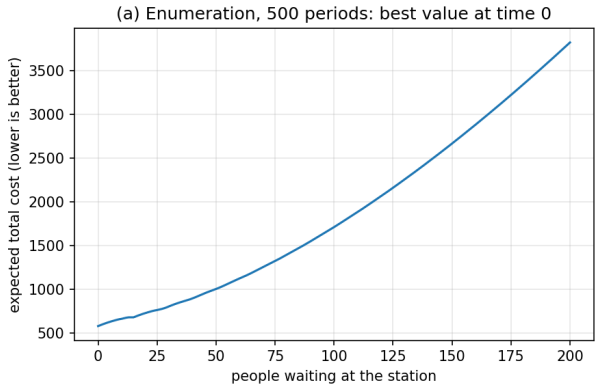


Figure 1. Enumeration: best expected cost at time 0 versus people waiting.

(b) Value iteration, no end date

The same step is repeated with no end date until the numbers stop changing (519 rounds). The curve is the same as in (a), which is expected: after 500 periods the far future is discounted to almost nothing, and the two differ by about 3 in a billion.

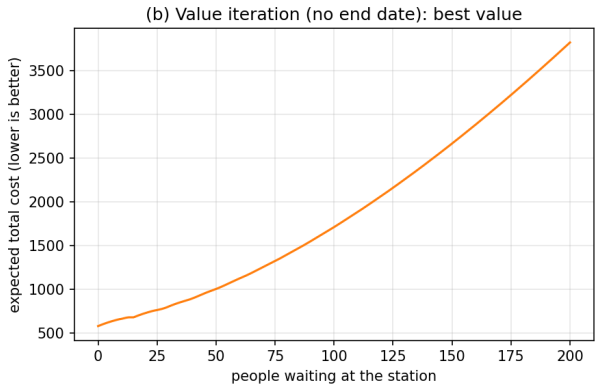


Figure 2. Value iteration: best expected cost versus people waiting.

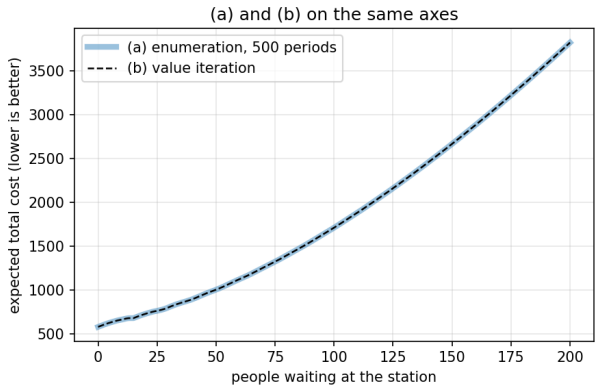


Figure 3. Figures 1 and 2 on the same axes. The two curves lie on top of each other.

(c) Policy iteration

Here we improve a plan instead of the costs. Starting from “never dispatch”, we measure how good the plan is, change it wherever another action is cheaper, and repeat. It settled after 6 rounds. The best plan is to wait while fewer than 13 people are waiting and to dispatch at 13 or more.

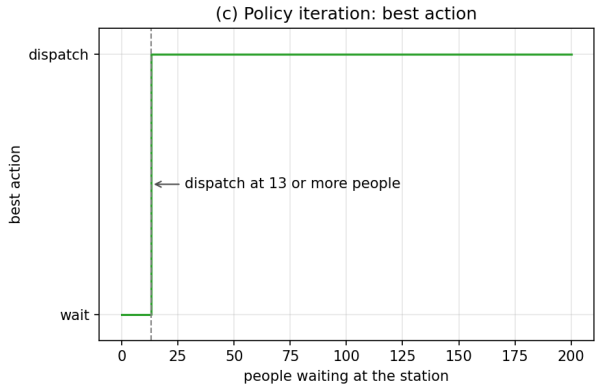


Figure 4. Policy iteration: best action versus people waiting.

Question 2: Edge cases and tests

To check the solution we looked at unusual situations (edge cases) and at changes to the problem where we can predict the answer in advance. All 23 checks pass. They are run with `python test_q1.py` and are grouped below. The check numbers match the ones printed by the program.

A. Ordinary situations

Case	What we expected	What we found
A1. Empty station (0 people)	Wait: a shuttle would cost 100 to carry nobody.	Best action is to wait.
A2. Dispatching with fewer than 15 people (4)	The shuttle takes everybody; we pay only the 100.	Cost 100; next period has 1 to 5 people.
A3. Does the 200-person limit matter? Try 300.	No change to the best actions.	Same actions up to 150 people; costs differ by 9e-11.
A4. Chances of the next situation add up to 100%	Exactly 100% in every situation.	Largest error 0. A deliberately damaged table is caught by the same check.
A5. A bigger shuttle	Wait for more people before sending it.	Shuttle of 5 / 15 / 30 dispatches at 5 / 13 / 17 people.

B. Edges of the station

Case	What we expected	What we found
B1. Exactly 15 people, then 16	15 empties the station; 16 leaves one behind.	Costs 100 and 102.
B2. Full station (200)	Dispatch, and waiting cannot push it above 200.	Best action is to dispatch; waiting stays at 200.
B3. Nearly full (199 and 198)	At 199 every arrival is turned away. At 198 only one fits.	From 199: always 200. From 198: 20% to 199, 80% to 200.
B4. Off-by-one	201 situations (0 to 200), not 200.	All tables are 201 wide and the graphs have 201 points.

C. Arrivals

Case	What we expected	What we found
C1. Nobody ever arrives, 10 waiting (worked by hand)	Waiting forever costs 20/(1-0.95) = 400; dispatching once costs 100. Cheapest: 100.	The program gives 100.000.
C2. Far more arrivals than the shuttle can carry (16 to 20 per period)	The queue never empties, so each person left costs 2/(1-0.95) = 40 forever. A dispatch (100) pays off from 100/40, so from 3 people. (Our first guess here was 15; working it through showed the guess was wrong, not the program.)	Dispatches for every number from 3 to 152.
C2b. Why does it stop at 152?	Near the limit the queue hits 200 anyway, so a dispatch saves little. The point should move if the limit moves.	Last dispatch at 152 (limit 200) and 252 (limit 300): an effect of the cap.

D. Extreme prices

Case	What we expected	What we found
D1a. Free shuttle	Dispatch whenever anyone is waiting.	Dispatches in 200 of 200 situations.
D1b. Free waiting	Never dispatch; total cost 0.	Never dispatches; cost 0.
D2. Enormous shuttle cost (1,000,000)	Never dispatch, even with a full station.	Never dispatches.

E. Time and discounting

Case	What we expected	What we found
E1. Very short horizon (worked by hand)	1 period left costs 2 x people. 2 periods left costs 3.9 x people + 5.7.	Both match the program exactly.
E2. Is 500 periods long enough? Compare 100, 200 and 500.	The answer settles down as the horizon grows.	Difference from the no-end-date answer: 4.0 / 0.024 / 0.000000003.

F. Checks on each method

Case	What we expected	What we found
F1. Enumeration: the starting point	Nothing left to pay after the last period; more periods left never costs less.	Value after period 500 is 0 everywhere; costs only grow with more periods.
F2. Value iteration: different starting guesses	Same answer from any guess.	All 0, all 1000 and random guesses agree to 4e-9.
F3. Policy iteration: improving and ties	Each round improves the plan. When all actions cost the same, it must not loop forever.	6 rounds, costs never went up. The all-ties problem stopped after 1 round.

G. Overall checks

Case	What we expected	What we found
G1. All three methods agree	Same costs and same plan.	Biggest gap 3e-9; identical plan.
G2. Policy iteration from a bad start (“always dispatch”)	Still finds the same best plan.	It does.
G3. Simulate the station 3000 times	The average cost matches the calculated cost.	Simulated 575.8 (±1.7) against calculated 576.7.